



**Program** sequence of binary instructions  
 CPU control unit → control signals

**datapath** ALU  
 Registers  
 Buses

**Computer** CPU, Main Memory, Interconnections, Input Output

distributed client-server, peer-to-peer  
 embedded low power, specialised cores  
 desktop  
 mobile

**CPU** fetches instructions, decodes instruction, performs instruction

**Microprocessor** all CPU components on 1 chip.

**The Fetch And Execute Cycle**

**Program Counter (PC)** next instruction address  
 Processor fetches instruction  
 Increment PC (unless told otherwise)  
**Instruction Register (IR)** instruction loaded into  
 Interpret instruction and perform action(s)

**Processor-memory** Clock Speed signal pulses/s e.g. GHz

**Processor-I/O**

**Data Processing**

**Control**

**Combination**

**Moore's Law** transistors on a single chip will double every year  
 higher density, smaller size, reduced power and cooling costs

**Arithmetic Logic Unit**

performs arithmetic and logical operations (decisions)

gateway between primary memory and secondary storage.

**Registers** small amounts of high-speed memory on chip to reduce need for main memory access

**User-visible:**

general purpose (AC)  
 condition code (flags)

**Control and status:**

program counter (PC)  
 instruction register (IR)

composed of logic gates that combine to perform:  
 ADD

ADD with CARRY

SUBTRACT

SUBTRACT with BORROW

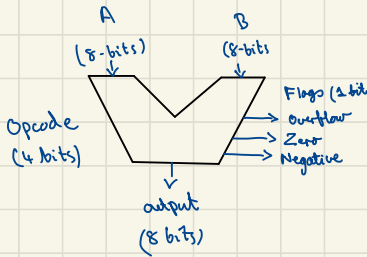
NEGATE

INCREMENT

DECREMENT

PASS THROUGH

**8-bit ALU**



memory address specifies a specific gated latch  
 to store or retrieve a bit from

decoder reads a binary memory address

**Control Unit**

Control unit registers & decoders

Sequencing logic

Central memory

**Optimisation**

**Pipelining** - staggered instruction/execution cycles

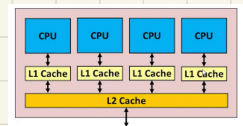
bad with branches  
 branch predictions

**Data Flow Analysis** - processor checks instruction dependencies  
 prioritises

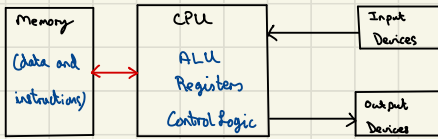
**Increase hardware speed** - shrink logic gates  
 power ↑

Increase size and speed of caches  
 Make DRAM wider

**Multiple Cores** large shared cache (L2)



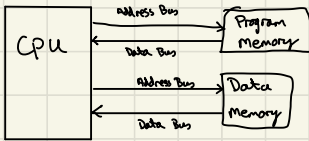
## Von Neumann Architecture



Von Neumann Bottleneck: all data had to go through CPU between memory and I/O

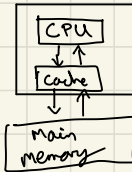
IAS Computer one general purpose register, Accumulator

## Harvard Architecture



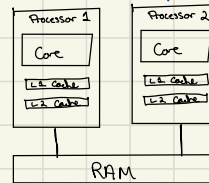
Simultaneous instruction and data fetching  
Convolutional

## Von Neumann with Caches



## Multicore Processors

cores run independently



L1 Cache 8-32 KB  
L2 Cache 1MB (external, connected to main)

Instruction Set all instructions understood by CPU

| Opcode | Operand Reference | Operand Reference |
|--------|-------------------|-------------------|
|--------|-------------------|-------------------|

## Reduced Instruction Set Computer (RISC)

small number of fundamental instructions

## Complex Instruction Set Computer (CISC)

large number of more complex instructions

CISC instructions take more clock cycles

more complex → more instructions → more power

Energy-sensitive devices e.g. mobile phones use RISC

## Instruction Type

|                     |                                                |
|---------------------|------------------------------------------------|
| Data Transfer       | Mov R1, R2                                     |
| Arithmetic          | ADD R1, R2                                     |
| Logical             | CMP R1, R2                                     |
| Conversion          | Convert integer to floating point              |
| Input/output        | move register content to graphics card         |
| System Control      | Reset CPU                                      |
| Transfer of Control | JMP #369                                       |
| Unconditional Jump  | loads the Program Counter with the JMP address |
| Conditional Jump    | JNZ #728 if true increment PC, false load #728 |

## Addressing Modes

- 1. Immediate operand is part of instruction no memory reference, fast limited use `LDR R1, 15`
- 2. Direct operand in address field limited address space `LDR R1, x`
- 3. Indirect operand in address in address nested, hence slow `ADD A`
- 4. Register operand is held in register no memory access needed, fast execution
- 5. Register Indirect operand is in memory cell pointed to by register
- 6. Displacement address field has base + displacement in register useful for consecutive memory cells
- 7. Stack operand is implicitly on top of stack

Stack push value, stack instruction, pop result  
used to pass arguments of a function call e.g. func(5, 7)

## Interrupt

interrupt normal flow/sequence of instructions  
important to prioritise certain function e.g. mouse pointer  
CPU's IRQ receives interrupt, saves all relevant registers (context saving)  
runs individual interrupt handler / interrupt service routine  
resumes after end of ISR (context restoring)

Simultaneous multiple interrupts  
disable interrupts  
further interrupts remain pending  
define priorities

## Buses PCI

address bus carries memory addresses from the processor to other components e.g. primary memory, I/O devices  
data bus carries data between processor and other components  
control bus carries control signals, clock pulses  
multiple > single bus

centralised v.s. distributed Arbitration

## I/O Module

Control, timing, error handling

CPU communication

Data buffering

CPU/Main Memory

I/O devices

I/O module

high data rate

variable

buffers

## I/O Techniques

Programmed

CPU has direct control

CPU request

wait bit

I/O performs, sets status bit

CPU periodically checks for completion

wastes CPU time

Interrupt Driven

CPU checks for cycle

requires active CPU  
transfer data limited

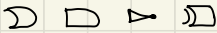
Direct Memory Access

intermediate between

takes over for a cycle

detached suspended twice  
(read/write)  
integrated suspended once

Logic Gates output signal from Boolean operation on input signals



Combinational Circuits sets of logic gates

Multiplexers control multiple inputs to a single output

Decoders

Adders

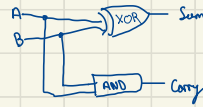
8-bit ripple carry adder

one half-adder

seven full-adders

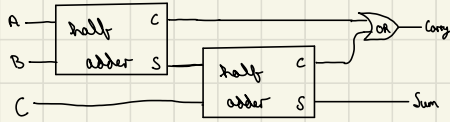
if 4th full-adder has a carry: overflow

half-adder:



| A | B | (Sum)<br>XOR | (Carry)<br>AND |
|---|---|--------------|----------------|
| 0 | 0 | 0            | 0              |
| 0 | 1 | 1            | 0              |
| 1 | 0 | 1            | 0              |
| 1 | 1 | 0            | 1              |

full-adder



| A | B | C | Carry | Sum |
|---|---|---|-------|-----|
| 0 | 0 | 0 | 0     | 0   |
| 0 | 0 | 1 | 0     | 1   |
| 0 | 1 | 0 | 0     | 1   |
| 0 | 1 | 1 | 0     | 0   |
| 1 | 0 | 0 | 0     | 1   |
| 1 | 0 | 1 | 0     | 0   |
| 1 | 1 | 0 | 1     | 0   |
| 1 | 1 | 1 | 1     | 1   |

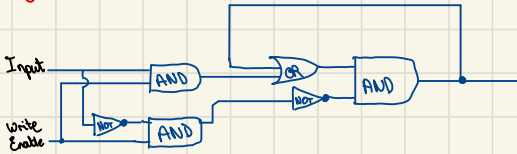
Sequential Circuit

output depends on input and current state

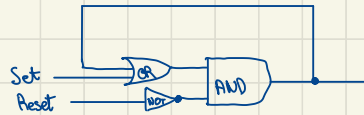
Flip-Flop / latch

can store a state  $\rightarrow$  one-bit memory

gated Latch



And-Or Latch



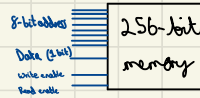
256 bits: 16x16 matrix of gated latches

DRAM

stored in capacitor, requires refreshing

simple, cheap

leaks charges



x8 can store 256 bytes

SRAM (Static Random Access Memory)

stored in switches/latches

no leak, faster

more expensive

Cache

CPU checks cache (word transfer)

if not present  $\rightarrow$  move from main to cache

(block transfer)

Read Only Memory

Programmable ROM

Erasable PROM

Electrically EPROM

"Flash" memory

Locality of Reference

references tend to cluster, more commonly accessed functions, loops

Write Through or Write Back

single or two-level

unified or split cache

Mapping  $C \ll M$

Direct (only 1)

Associative (any)

Set Associative (any in range)

Search Recently Used  
First in first out  
Search frequently used  
Random

NVRAM

Virtual Memory

Storage (non-volatile memory)

paper punch cards

delay line memory

|            |                      |                 |                   |                |             |
|------------|----------------------|-----------------|-------------------|----------------|-------------|
| mercury →  | magnetic core →      | magnetic tape → | hard disk drive → | optical disk → | solid state |
| pulses     | 1c per bit           | 10Kb            | 5mb               |                |             |
| sequential | random access memory | durable         | thin              |                |             |
|            |                      | cheap           | more storage      |                |             |
|            |                      | compact         |                   |                |             |
|            |                      | slow retrieval  |                   |                |             |

floppy disk

|             |               |
|-------------|---------------|
| Sequential  | magnetic tape |
| Direct      | hard disk     |
| Random      | RAM           |
| Associative | Cache         |

**Memory Hierarchy** a combination of different types of memory to balance capacity, speed and cost

## Magnetic Disk

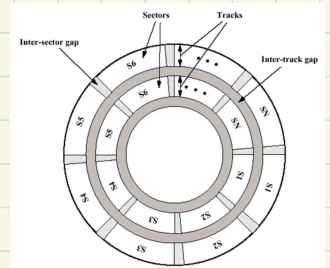
- circular platter with magnetisable substrate
- tracks (concentric rings) → inter-track gap
- moveable read/write head → sectors (minimum block size)
- smaller head → more tracks
  - ↳ enclosed
  - ↳ more robust
- double-sided (two heads), cylinder (multiple platters)
- positioning ∴ slow

$$\text{Access Time} = \text{Seek} + \text{Latency (rotational)}$$
$$= \text{Direct} + \text{Sequential Access}$$

## Format Disk

delimit disk areas: mark tracks and sectors

- Constant Angular Velocity



## Solid State Drive

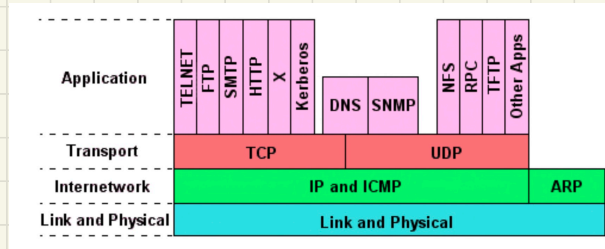
integrated circuits, Flash ROM  
faster, no moving parts

Read

# Networking

## Open System Interdia (OSI)

Application  
presentation  
Session  
Transport  
Network  
Data Link  
Physical



## Data Link Layer (Medium Access Sublayer)

Logical Link Control (LLC) frame structure, address schemes, control information

Media Access Control Address (MAC) how a device can access the network

resolves contention for shared media  
single channel, multiple hosts

Carrier Sense Multiple Access

1-persistent  
non-persistent  
P-persistent (rotated)



BitMap Contention Slots

## Network Layer (routing packets)

Static (Offline) Routing

Shortest Path  
Flow Based  
Flooding  
Dijkstra's, Breadth-First Search

Dynamic Routing

Distance Vector  
Link State  
router tables

Network Address Translation

Private IP → public IP address

Dynamic Host Configuration Protocol Address Resolution Protocol (ARP)

IP assignment and renewal

IP → MAC address



Hardware  
repeater amplify signal  
hub multipoint repeater  
bridge / switch multipoint repeater, traffic filter  
gateway regulate dissimilar protocols

## Transport Layer

User Datagram Protocol (UDP)



port data to right application  
checksum

Transmission Control Protocol (TCP)

port  
checksum  
sequential packet numbers  
acknowledgement packet error

## Application Layer

Domain Name System (DNS)

top level domains  
second level  
sub-domain

Telnet

Secure Shell (SSH)

File Transfer Protocol (FTP)

Hyperfast Transfer Protocol (HTIP)

## Concurrent processes

Monoware

Indirectly Aware

Directly Aware

## Memory Management Unit

Partitioning

Pages

Virtual Memory

## Files and File Management

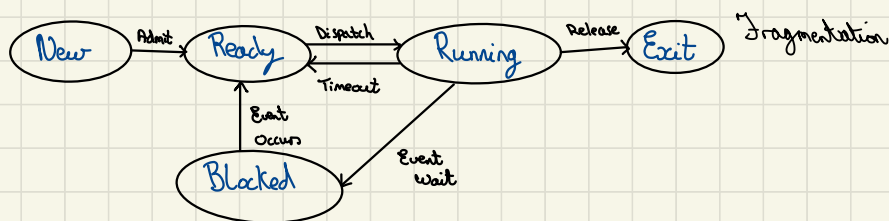
### File Format

metadata in header

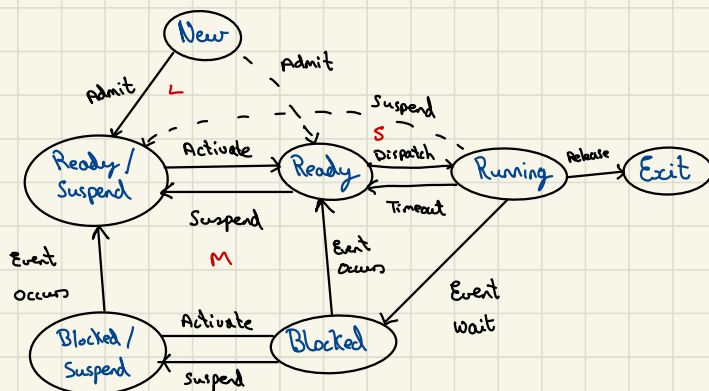
### Directory File

root directory

blocks,  
slack space



Fragmentation



## Process Scheduling

First Come First served unpredictable easy

Shortest Process Next long process starvation, minimise average time.

Shortest Remaining Time overhead, penalises long fast for short

Round Robin

## Semaphore

wait(S):

$S.count := S.count - 1$

If  $S.count < 0$

then begin

Place P in s.queue

Block this P

end

signal(S):

$S.count := S.count + 1$

If  $S.count <= 0$

then begin

Remove P from s.queue

Place P on ready list